

A Journal Club-Style Review of: *One-second coherence for a single electron spin coupled to a multi-qubit nuclear-spin environment*

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I. INTRODUCTION

In any application of quantum computing, maximizing the coherence time of a qubit is critical for its usefulness; the longer the qubit accurately maintains its state, the easier it is for that qubit to undergo gate operations, become entangled with other qubits, and for longer quantum algorithms to be used. In particular, solid-state qubits such as NV (Nitrogen Vacancy) centers have been a target for many researchers who are now looking to extend its coherence to the scale of seconds. By combining several previously-successful techniques used to characterize the spin environment surrounding the NV center, this research group achieved a coherence time of 1.46 seconds for a single qubit [1]. Their work has established a standard practice for future NV center research and may have further applications in other fields such as quantum encryption.

II. BACKGROUND AND CONTEXT

Prior to this paper, there had been several significant developments in the world of NV center electron spin qubits which these researchers made particular use of. As we previously discussed in class, the main issue for these qubits center around their T_2 related phase decoherence, which is caused by interactions with other nearby spins: namely, those from ^{13}C atoms that naturally occur even in the most pure diamond samples (at a rate of 1.1%). When gate operations are being applied to the qubit, sometimes driving frequencies can be chosen which cause unwanted coupling to these ^{13}C spins, causing the electron spin to act unexpectedly and thus losing coherence. There are two main cases of this issue which previous papers have begun to address, one being when the electron spin interacts with isolated ^{13}C spins [2], and another when the electron spin interacts with ^{13}C - ^{13}C spin pairs [3].

For a long time, research groups working with NV centers have been aware of these ^{13}C spins, and begun to characterize their nature based on their proximity to the electron spin. In 2006, a research group found that these isolated ^{13}C spins which are located close to the electron spin have a very strong hyperfine interaction, causing sharp peaks of state decoherence to occur over a short timescale (on the order of μs) [4]. Later groups were able to take this information about nearby ^{13}C spins and use it as a prediction tool, allowing them to effectively control these unwanted spins and recover the lost coherence [2].

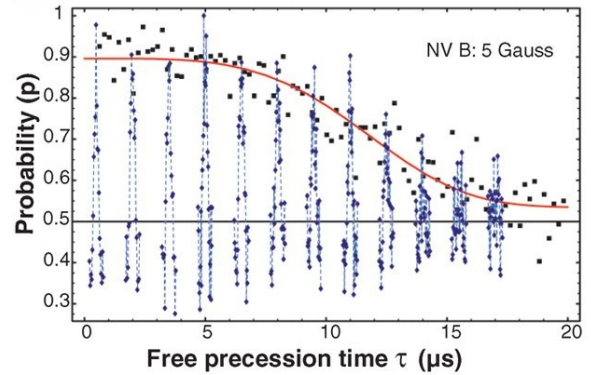


FIG. 1. A graph of the coherence of a electron spin qubit, with sharp decoherence peaks due to interactions with isolated ^{13}C spins. Modified from [4].

While mapping these isolated ^{13}C spins was a major step towards improving the T_2 coherence of the NV center qubit, there were still longer dips in state fidelity that needed to be addressed. Around the same time, another research group focused on characterizing a single ^{13}C - ^{13}C spin pair near an NV center, which result in decoherence dips on the scale of hundreds of μs [3]. Through a similar (although more complex) version of the quantum sensing process previously used, this spin pair was able to be successfully located by testing a variety of driving frequencies. Another group was also able to identify over 20,000 of these spin pairs in a special ^{13}C depleted sample using many dynamical decoupling sequences [5]. As we will see in the specific paper being reviewed, combining the processes for sensing both isolated ^{13}C spins and coupled ^{13}C spin pairs will be particularly useful in preserving electron spin coherence.

Previous works have also offered insight into which dynamical decoupling sequences are the most effective at maintaining long qubit coherence times (depending on the experimental setup). In class, we studied the Hahn Echo sequence in detail, which is a simple, one-axis dynamical decoupling sequence designed to regain coherence by "flipping" the qubit state periodically (applying repeated π_y pulses), counteracting detuning factors that are roughly constant across two sets of pulses (2τ). In more recent years, a variety of dynamical decoupling sequences have been tested specifically on NV centers, including those with two-axis pulses, aperiodic pulses, and symmetrical or concatenated variants of these sequences [6]. Of these, the two-axis, symmetrical XY8 pulse sequence ($\pi_x - \pi_y - \pi_x - \pi_y$ followed by $\pi_y - \pi_x - \pi_y - \pi_x$) is recommended as the most robust option for maintaining long coherence, as it is relatively easily to implement experimentally while mitigating the effects of large pulse errors (which can occur when attempting to maximally reduce τ , as we will later see) [6].

III. ACCOMPLISHMENTS OF THIS PAPER

In this paper, the research team aimed to increase the coherence time for a single electron spin in an NV center. To accomplish this, they first characterized the NV center's surrounding environment of ^{13}C atoms. By combining several different quantum sensing tests and mathematical models (versions of those seen in [2] and [3]), the group was able to determine that near this NV center, there are 6 ^{13}C - ^{13}C spin pairs and 7 isolated ^{13}C spins (shown in figure 2). By modeling the nature of these ^{13}C atoms and the surrounding bath of distant ^{13}C atoms (due to the 1.1% natural occurrence of ^{13}C), the team is able to predict which frequencies will drive unwanted interactions, and avoid any of these transitions which would cause the qubit state to decohere [1].

Once the environment was well-established, the research group analyzed both T_1 and T_2 related decoherence of the NV center. First, the longitudinal relaxation (θ shift) of the electron spin was tested by repeatedly initializing the qubit into its eigenstates $|m_s = 0, -1, 1\rangle$ and measuring it after certain evolution times. The group found that their T_1 coherence time is on the order of 10^3 seconds (illustrated in figure 3), demonstrating that the energy-related longitudinal relaxation is far from being a limiting factor for qubit coherence [1].

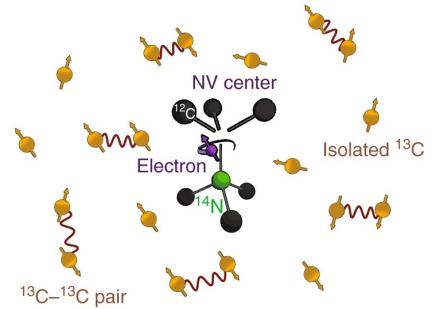


FIG. 2. A model of the NV center, surrounded by 6 coupled ^{13}C spins and 7 isolated ^{13}C spins. Modified from [1].

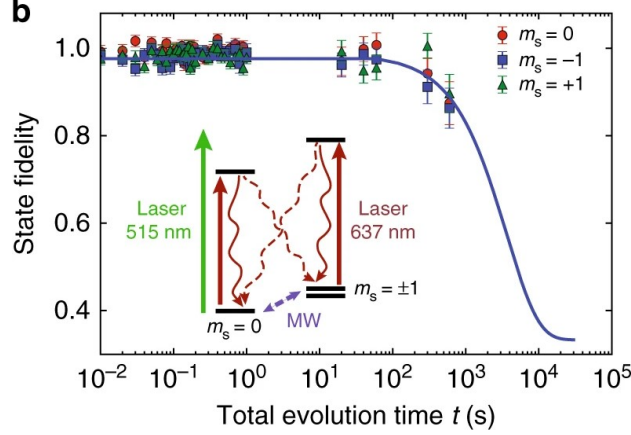


FIG. 3. A graph of the coherence of the qubit after being initialized to its eigenstates. Modified from [1].

The rest of the paper focuses on mitigating the T_2 related decoherence of the NV center, primarily caused by the ^{13}C spins previously modeled. To do this, the group employs an XY8 dynamical decoupling scheme. A time spacing 2τ can be chosen between between flipping operations (a mix of π_x and π_y). A shorter τ helps to mitigate the effects of the surrounding ^{13}C bath, but τ is also limited by its timing resolution (appx. 1 ns). By tuning τ to land on a harmonic of the ^{13}C Larmor frequency (which should theoretically minimize decoupling) while also factoring in the effects above, the optimal results were achieved [1].

Figure 4 below highlights the state fidelity (or coherence) of the qubit after being prepared in the $|+x\rangle$ state. Each dot represents a measured state fidelity recorded for the qubit over a certain interval of time, which was repeated over many time intervals to produce fitted curves. The thick lines assume an ideal environment, while the thin lines factor in the 6 coupled ^{13}C spins. Each set of colors represent a different N value, corresponding to the number of times the XY8 scheme was repeated. Lower N values mean that τ is longer, while greater N values mean that τ is shorter. As shown in the plot, the greater N values enable the longest coherence times, but the limited timing resolution causes the initial state fidelity values to drop off [1].

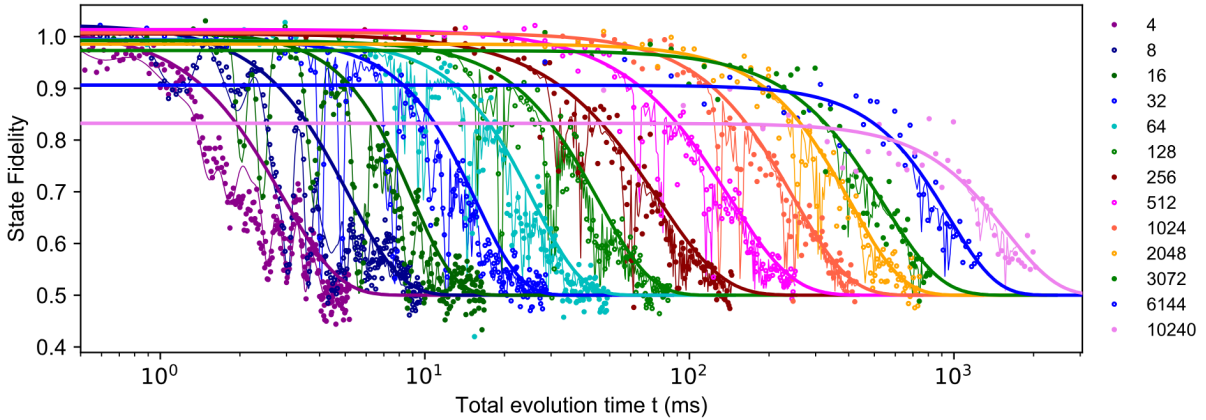


FIG. 4. Overlaid plots of qubit coherence initialized in the $|+x\rangle$ state. Each colored line represents a different N value, ranging from $N = 4$ to $N = 10,240$. Modified from [1].

Combining all of their data across the six cardinal quantum states, the group was able to assert that by selecting $N = 10,240$, they can preserve any quantum state for up to 1.46 seconds (with a fidelity above the standard $2/3$ classical limit). By factoring both the paired and isolated ^{13}C spins into their model, the group was able to recover more information from their qubit compared to previous studies, and achieve the longest coherence time for solid-state electron qubits to date [1].

IV. METHODS AND SAMPLE SELECTION

As common with other solid-state qubits, this study conducted its experiments at a cold 3.7 K, which can be accomplished with commercial products such as Montana Instruments' Cryostation, as used in this case [1]. These devices often have quite small spaces available for samples (usually only hold thin chips no more than 20 centimeters in width or length), but this is all that is needed to house the sample containing the NV center [7].

The sample used in this experiment is a type IIa chemical vapor deposition (CVD) grown diamond, which is a common choice among NV center qubits (see [2, 8]). In the CVD growth process, an initial small "seed" of pure diamond is placed in a chamber which is then flooded with a reactant gas [9, 10]. This gas is then adsorbed by (deposited onto) the surface of the diamond seed, and over time, a thin film of diamond is built-up on top of the initial seed [9, 10]. This diamond can then be analyzed for imperfections (by being separated into regions based on dislocations which propagated from the initial seed) and cut as needed [11]. Although the diamond seed is synthetically grown from a different process (high-pressure-high-temperature, or HPHT), CVD offers a much more consistent crystalline structure with a low amount of impurities.

The "type IIa" of the diamond sample simply refers to its purity, containing only very trace amounts of nitrogen or boron impurities. In order to grow the type IIa CVD diamond, a chain of other, less-pure synthetic diamonds must first be grown through the HPHT process which then are cut to become seeds [11]. Although the process is lengthy, type IIa diamonds grown this way can have a dislocation density of under 400 cm^{-2} near its center, which makes them ideal samples for qubits [11].

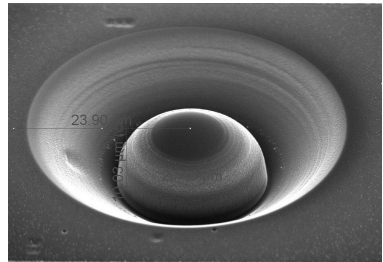


FIG. 5. An image of a solid immersion lens (SIL), modified from [8].

In this experiment, a solid-immersion lens (SIL) was also created in the sample above the studied NV center. Using a carefully-focused laser, a small hemisphere can be carved into the outer surface of the diamond, spanning only tens of microns in diameter (shown in figure 5). In similar experiments, this has helped to greatly increase the fluorescence measured by the detectors (200% more photon detections or greater) [8]. In addition, a layer of aluminum-oxide anti-reflection coating was added to the surface of the sample to further improve photon detection and limit unintended photon recordings [1]. This careful selection of previously-successful technologies enabled this research group to get clear data for their experiments, boosting the effectiveness of the other more technical experimental decisions.

V. IMPACT OF WORK

The most immediate impact of this research is the extended coherence time for any single electron spin qubit; the 1.46 second benchmark set by this paper nearly tripled the previous best coherence time achieved using a silicon-based qubit [12]. However, the benefits of breaking the one-second barrier with NV centers (and the methods used to do so) go beyond just extending coherence time. For example, later researchers were able to make use of the pulse-spacing (τ tuning) technique used in this paper to achieve five second coherence times in another type of silicon-based qubit [13]. This paper was also the first to factor in multiple ^{13}C - ^{13}C spin pairs, setting a standard for future studies in extending coherence times in solid-state qubits [1]. This advanced modeling of the NV center’s environment also has potential applications in quantum sensing (expanding on sensing techniques used in previous studies like [5]).

The extension of solid-state qubit coherence times to the magnitude of seconds also has significant implications for quantum networks and quantum encryption. Previous studies have shown that if roughly half-second coherence times could be achieved in NV center qubits, two remote quantum nodes could be entangled using photons above the critical link efficiency required for effective quantum communication [14]. Although the realization of larger quantum networks still requires other issues to be solved (such as the need for effective quantum repeaters and improved photon-detection efficiency), NV centers appear to be a prime candidate for these systems [14].

VI. CONCLUSION

By combining previous techniques used to characterize and mitigate the effects of ^{13}C spins surrounding an NV center, this research group managed to significantly improve the phase coherence of their qubit and demonstrated that their work is easily applicable to other solid-state qubit research. Careful selection of a dynamical decoupling scheme (XY8) and diamond sample (grown from CVD) alongside other experimental decisions (including the solid-immersion lens) allowed this team to capitalize on their novel work. Later groups have been able to successfully emulate the quantum sensing protocols used in this paper to further the coherence time of other solid-state qubits, providing further proof-of-concept for their potential use in quantum encryption.

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